

REVIEW

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A review of autonomous and low cost sensor systems for surface water quality monitoring with implications for local decision making

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Abstract

Preserving water quality has become a critical priority due to the increasing pressure on water resources from population growth, production activities and climate change. This article reviews recent studies on water quality monitoring in lakes, rivers and reservoirs, with a focus on lessons relevant to local decision-making. Using a structured literature review approach, we synthesised and interpreted research on water quality monitoring systems used in aquatic ecosystems affected by eutrophication and turbidity. Particular attention was given to turbidity sensors and autonomous monitoring approaches. The findings show that traditional manual quality measurements continue to coexist with autonomous monitoring systems, each contributing different strengths to water quality assessment. However, studies on low-cost sensors and permanent autonomous measurement systems remain limited, likely because of other research priorities. The limited number of published studies suggests either that these systems remain uncommon in practice or that their use is insufficiently documented. Current monitoring efforts largely rely on field manual sampling carried out by specialised technical personnel employed in state services or research centres. Looking ahead, it appears promising to complement current methods by increasingly integrating low-cost autonomous sensor networks and expanding opportunities for citizen participation. Such developments could support the creation of real-time information systems that facilitate local decision-making.

Keywords Water monitoring, Lakes, Rivers, Autonomous sensors, Turbidity, Eutrophication

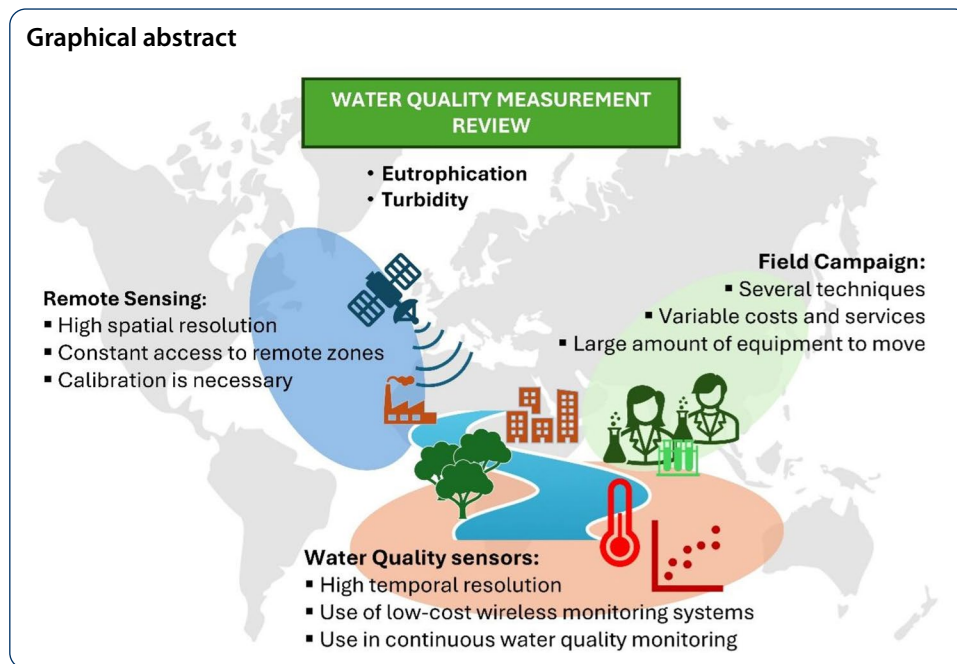
1 Introduction

1.1 Water quality: a global challenge

Ensuring water quality is one of the major environmental and societal challenges of the twenty-first century. Water quality plays a crucial role in public health and ecosystem conservation [28]. It is also central to sustainable development, as access to safe and reliable water supports economic growth and social well-being [10]. However, progress towards achieving the United Nations Sustainable Development Goal on clean water



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and sanitation remain insufficient, and improvements in water supply quality continue to fall short of global targets [99]. This challenge is particularly evident in Latin America and the Caribbean, where water quality continues to deteriorate due to combined effects of climate change and socioeconomic pressures [98]. The risk of water shortages due to pollution is a growing concern [104], exacerbated by extreme hydroclimatic events that increase the likelihood of droughts and degrade water quality [29]. Human activities also contribute to pollution along entire river systems, from upstream to downstream [44]. The two main threats identified for freshwater bodies are nutrient pollution, which leads from eutrophic to hypertrophic conditions and toxic stress from chemical pollution [97]. The World Bank has warned of the urgent need to address water quality issues before they escalate [23]. Over the past decade, water-related conflicts have increased worldwide, [45], and social movements advocating for the protection of water resources have also emerged [100].

1.2 Monitoring water quality to reduce exposure to pollution and its effects

Limited water quality information in low-income countries remains a major challenge that must be addressed to enable informed decision-making [99]. Beyond technological innovation, the evolution of water-quality monitoring systems reflects a broader transition in environmental science towards data-intensive observation frameworks that support anticipatory and evidence-based water governance. Improved knowledge of water quality enhances the capacity to anticipate health risks. Poor-quality or contaminated water affects mental and psychological well-being as much as physical health [28], increasing the incidence of infectious diseases, which are often linked to microbiological contamination. Contaminated water can also cause organ damage and cancer due to exposure to inorganic and organic contaminants [19].

Effective water purification strategies are essential for maintaining public health while supporting agriculture and conserving biodiversity. Yet, as water challenges vary

widely across regions, these strategies are most appropriate when informed by both local knowledge and expertise [28]. Monitoring programmes and water-resource management are crucial for reducing the risk of microbial and chemical contamination of drinking water, often driven by land-use practices in river basins [106].

The World Health Organisation has also underscored the responsibility of governments to fund and coordinate national water quality monitoring programmes by promoting initiatives that encourage communities to participate in citizen science, known as community-based monitoring [107]. Developing citizen science initiatives involving researchers, citizens and public authorities for long-term water quality monitoring could help fill information gaps, reduce waterbodies degradation and raise public awareness [20].

1.3 Emerging WQMS: examples from Latin America

With abundant surface and groundwater resources, Latin America has the highest per capita water availability in the world , [16]. However, land use strongly influences water quality, particularly through increasing urbanisation [91]. Numerous cases of pollution have been documented across the region, linked to agricultural activities [16, 35], industrial discharges [33] and domestic wastewater [16, 76]. Pollution remains difficult to control due to high financial costs, inadequate wastewater-treatment systems [44] and insufficient water quality monitoring programmes , [16].

Many Latin American countries operate permanent public WQMS which focus on major surface waters or key watersheds. Nevertheless, these data often consist of irregular point-in-time measurements rather than real-time observations. Government agencies often manage online platforms that provide public access to this data, such as the Water Quality Information System (SINA) in Mexico [22] and the National Information System for Integrated Water Resource Management (SINIGIRH) in Costa Rica [57] in Central America. Other examples include the online monitoring system of Brazil's National Water and Basic Sanitation Agency [3], the Environmental Information Centre (CIAM) in Argentina [83] and the information systems of the General Directorate of Water in Chile [26].

However, within that world region, these measures remain insufficient to address persistent water quality problems. The uneven distribution of the thousand monitoring stations limits the availability of comprehensive data for real-time decision-making [66]. The implementation of on-site sensor networks and the establishment of additional monitoring stations will be essential to achieving adequate territorial coverage. Another important consideration is that citizen monitoring remains underrepresented and undervalued in water quality monitoring systems (WQMS). These initiatives aim to track changes in drinking water quality and safeguard the surrounding aquatic ecosystems. However, its range of analysis methods can be adapted to local environmental and socio-economic conditions, complementing official hydrological data monitoring by facilitating the detection of pollution and environmental changes in areas lacking formal monitoring [13]. Similar to the challenges observed in Latin America, a research gap remains regarding non-interconnected lakes in emerging economies [55]. Where autonomous monitoring is still limited, the combination of multivariate analysis with GIS mapping is essential to identify critical pollution areas and guide local management strategies.

1.4 Turbidity and eutrophication as main factors of disturbance

Turbidity is a key water quality parameter and is widely used as an indicator of trophic status and hydrosystem productivity. It may reflect the presence of suspended particles in the water column, arising from both organic matters [12] and inorganic suspended solids [36]. Elevated turbidity reduces light penetration through increased light scattering, thereby affecting aquatic ecosystem functioning [82, 85].

Turbidity exhibits strong seasonal and interannual variability [1, 43, 79, 96], driven by meteorological conditions that affect biological, and sediment transport processes [36]. Depending on local climatic conditions, turbidity peaks may occur during summer or wetter autumn/winter periods [1, 39, 50, 51, 102]. Extreme events, including storms, strong winds, typhoons and monsoons, can substantially increase sediment resuspension and run-off, often contributing disproportionately to annual sediment loads [1, 42, 109]. In addition, the relative contribution of inorganic particles and phytoplankton varies among systems, with algal blooms commonly intensifying turbidity during warm periods [53]. In shallow lentic systems, turbidity is closely associated with nutrient concentrations particularly total phosphate and nitrogen, highlighting the critical role of nutrient loading in determining water quality [55].

Both natural and anthropogenic factors influence turbidity and nutrient dynamics. Catchment characteristics, including topography, soil properties and land-use practices, regulate erosion and sediment delivery [38, 54], while hydrodynamic conditions, water depth and sediment resuspension control turbidity within lakes and rivers [38, 54]. At the lake or river scale, turbidity is affected by water column depth [43, 68], hydrodynamic conditions [43, 82], bed and bank erosion, sediment deposition and resuspension. Within lakes, turbidity differs between the fluvial zone [78] and deeper or bottom regions. Human activities such as agriculture, urbanisation, industry and aquaculture increase sediment and nutrient inputs, particularly nitrogen and phosphorus, through runoff and tributary inflows [60, 70, 74, 92, 103, 108, 112]. Nitrogen pollution is strongly associated with agricultural activities in catchment areas, primarily as a result of excessive fertiliser application [1, 56, 60, 108]. Tributaries redistribute pollutants from urban and agricultural areas, resulting in cumulative downstream impacts [43, 52, 108, 115]. These inputs stimulate algal growth, increase chlorophyll-a concentrations and can reduce dissolved oxygen levels, ultimately degrading water quality and ecosystem health [48, 84].

The combined effects of climate change and nutrient enrichment contribute to eutrophication, one of the main causes of degradation in aquatic ecosystems in lakes and wetlands [18, 25, 88] [39, 48, 50, 51, 60, 102, 108]. Rising temperatures accelerate algal productivity and may concentrate dissolved constituents through increased evaporation, exacerbating water quality degradation. [48, 55, 60, 78, 103]. As a result, eutrophication threatens key ecosystem services such as recreation, irrigation and drinking water provision [89]. These interactions underline the need for monitoring systems capable of capturing the temporal and spatial variability of turbidity and nutrient dynamics to support effective water resource management.

1.5 Objective, hypothesis and article outline

As environmental monitoring rapidly transitions toward continuous, data-intensive observation and advanced analytical approaches that support decision-making,

a synthesis is needed to understand how these advances are reshaping both scientific understanding and governance. This article aims to synthesise the literature on the role of autonomous systems within water quality monitoring systems used to address environmental risks in rivers, lakes and reservoirs. Our research focuses on surface waters affected by runoff, erosion, and acute eutrophication, and as such does not include groundwater. Indeed, the cost-effective, autonomous optical turbidity sensors and telemetry equipment presented in this article are designed for surface deployment to capture high-frequency spatial and temporal dynamics induced by weather events and land-use practices. While groundwater quality is a critical component of water resource management, its monitoring presents entirely different hydrogeological dynamics, access limitations, and sensor deployment challenges, which fall outside the technological framework of the affordable surface water telemetry examined here.

We prioritised turbidity as a proxy variable and required these keywords to ensure comparability across systems. Our review was guided by two hypotheses. First, preferred methods for monitoring the quality of continental would rely primarily on time-consuming, labour-intensive field sampling campaigns or, alternatively, satellite imagery. And second, the adoption of low-cost, autonomous and easy-to-use local technologies would remain relatively uncommon, creating gaps in real-time data availability for local decision-makers and concerned communities. In this context, local monitoring systems could serve as powerful tools for community-level decision-making and may be readily adopted by local populations. We paid attention to the most used tools for assessing water quality, seeking to address the challenges associated with developing and implementing low-cost monitoring systems.

1.6 The novelty of the review

This paper highlights the emerging use of low-cost, autonomous sensor networks as a rapidly evolving approach to water quality monitoring in high-resource countries. In contrast to traditional systems that rely on manual, point-in-time sampling, these technologies generate continuous, real-time data, addressing longstanding limitations associated with irregular or delayed information. This enables early detection of quick or short-lived disturbances such as turbidity and eutrophication, which are particularly relevant in the region. Moreover, the reviewed studies point to the potential for stronger community participation in water monitoring, helping close persistent gaps in timely and actionable information for decision-making, in lower income countries, particularly in Latin America. Beyond their technical advantages, these systems represent a shift toward high-frequency environmental observation capable of capturing transient processes that traditional sampling often overlooks. This transition has important implications for hydrological modelling, early-warning capacity and adaptive water governance, particularly in regions where delayed information constrains timely decision-making, reinforcing the transition from reactive to anticipatory water management.

2 Material and methods

In order to review and synthesise the literature on the subject, we followed the PRISMA reporting guidelines, applying its procedural steps, flowchart and checklist [58, 59].¹ The article selection process is summarised in Table SI-1 and Fig. 1.

2.1 Literature search

In October 2025, scientific articles were retrieved from the Web of Science (WoS) database, selected for its extensive peer-reviewed literature, multidisciplinary coverage and advanced filtering tools. We used keyword-based search equations:

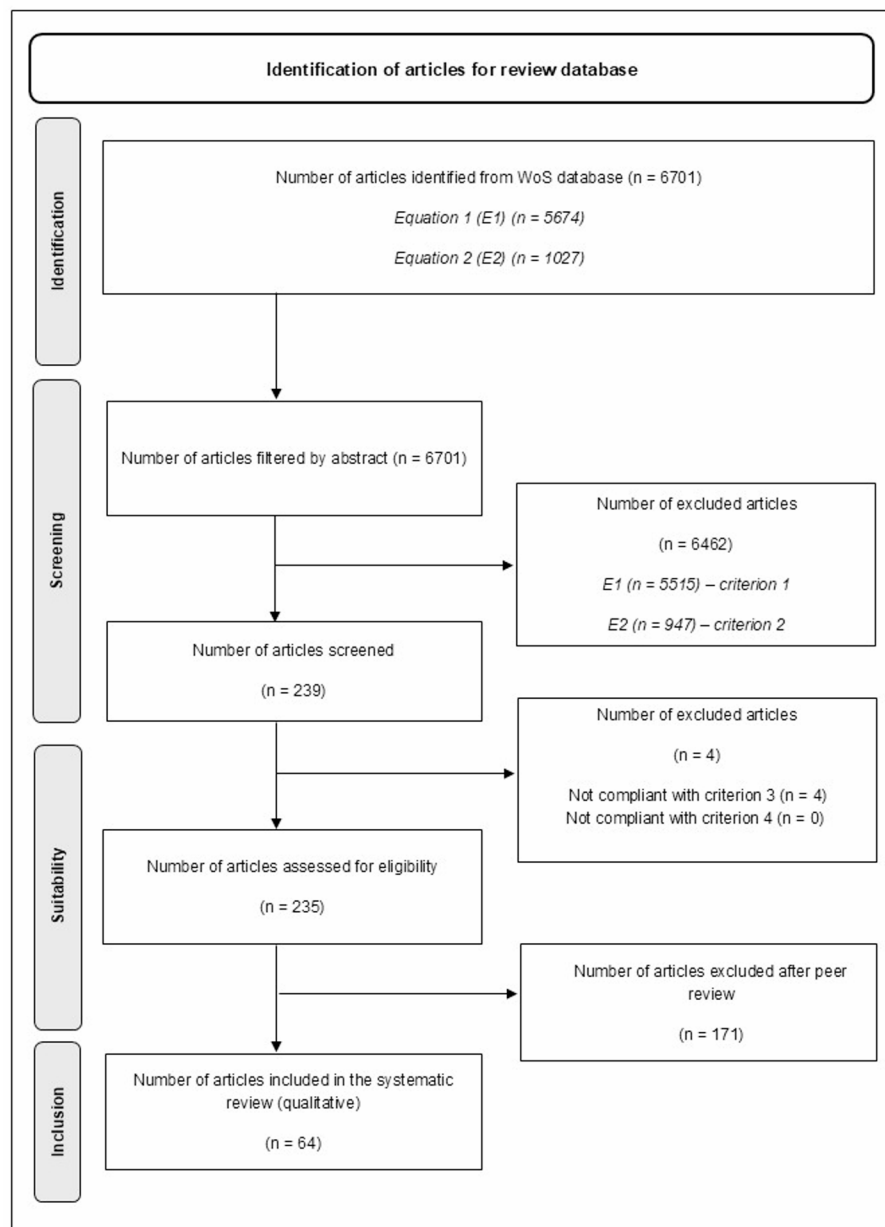


Fig. 1 Stages of the literature review process. Based on the PRISMA 2020 flowchart

¹ <https://www.prisma-statement.org/prisma-2020>

Equation 1= (lake AND “water quality” AND (eutrophication OR turbidity) AND monitoring AND sensor AND (“low cost” OR low-cost).

Equation 2= (turbidity AND “hydrological modelling” AND (catchment OR basin) AND sensor AND monitoring NOT lake).

Each search query was applied to all fields (topic, title, publication titles, abstract and keywords) and a filter was applied to include only articles. While no timespan filter was applied, the initial search returned 6,710 articles.

2.2 Inclusion and exclusion criteria

Due to the large number of results, inclusion and exclusion criteria were applied to obtain a more focused dataset.

The inclusion criteria were:

- i) For Eq. 1: articles had to contain the keywords ‘lake’, ‘quality’ and ‘turbidity’ (criterion 1).
- ii) For Eq. 2: articles had to contain the keywords ‘monitoring’, ‘sensors’ and ‘quality’ (criterion 2).

The exclusion criteria were:

- iii) Articles not written in English or Spanish (criterion 3). The language criterion applied to screened journal articles; background references in other languages were retained where relevant.
- iv) Duplicate articles were removed from the list (criterion 4).

These inclusion and exclusion criteria were subsequently automated using a custom R script. The script was designed to cross-reference target keywords within article titles and abstracts, which had been retrieved from the Web of Science (WoS) database using its built-in export functionality. The dataset was exported in CSV format for subsequent analysis. Application of these criteria reduced the dataset to 236 articles.

2.3 Screening procedure

The abstracts were screened independently by two of the listed authors (one from engineering, one from social sciences, both specialised in water studies), who coded each abstract. Articles were classified as:

- i) Clearly related to the topic
- ii) Potentially related; to be discussed
- iii) Unrelated to the topic

Articles coded ‘i’ by both reviewers were retained, while those coded ‘iii’ by both were discarded. For articles coded “ii”, inclusion or exclusion was determined through collaborative discussion between the reviewers. Reasons for exclusion included: overly general approaches, lack of remote-sensing or machine-learning components or WQ monitoring based solely on algae or fish sampling. A total of 172 articles were discarded, leaving 64 for analysis (see Fig. 1).

2.4 Analysis

We applied thematic analysis [64] using article titles and abstracts to classify the selected articles into inductively constructed categories aligned with the review hypotheses, emphasising water quality monitoring in hydrosystems for real-time decision-making. The detailed list of the 64 articles analysed can be found in Table SI-2.

3 Results. Water quality monitoring systems: from manual sampling to autonomous techniques. Benefits and prospects

In many countries, water authorities are responsible for ensuring the safety of water use and consumption, including maintaining standard biological and chemical levels in aquatic ecosystems [1]. To assess the condition of surface waters, monitoring is fundamental for identifying cycles and changes in water quality. As various studies suggest, these cycles—observed in rivers, estuaries, lakes, reservoirs and lagoons—are influenced by both natural factors and anthropogenic pressures [1].

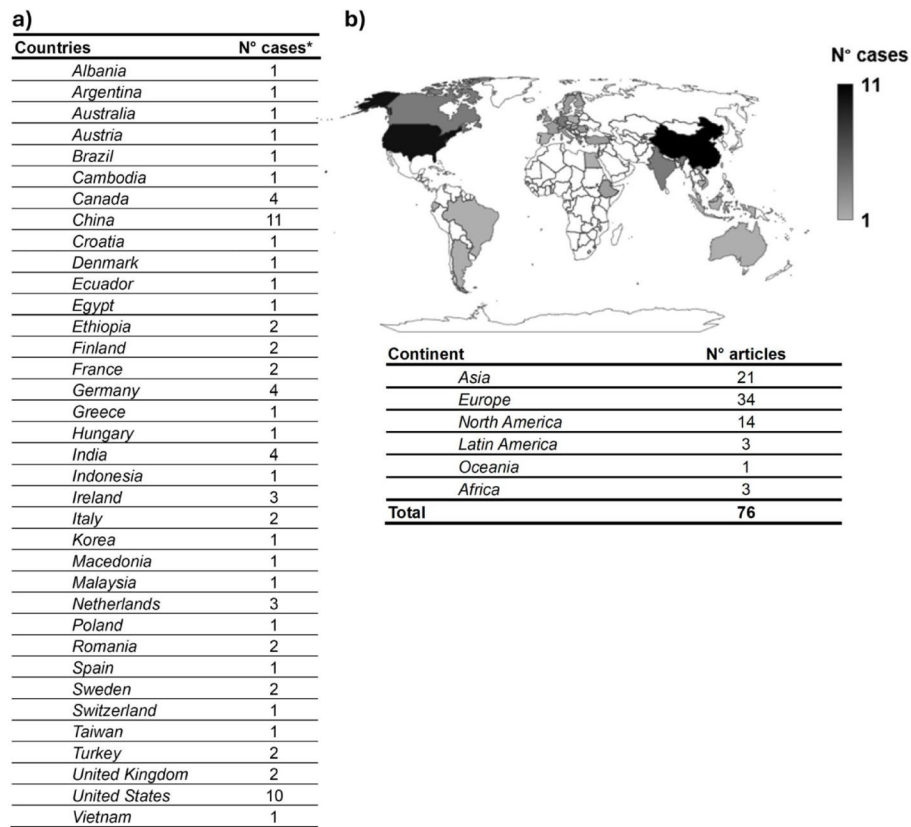
Over time, water quality monitoring has become increasingly technical and diversified. Methods range from manual sampling [41] to automated approaches, temporary campaigns [49] to continuous monitoring [62], and small-case analyses [49] to extensive basin-level assessments [87]. Given this diversity and the growing need for reliable data, analysing existing water quality measurement methods is particularly important for resource-constrained regions.

3.1 Bibliometric and geographical distribution

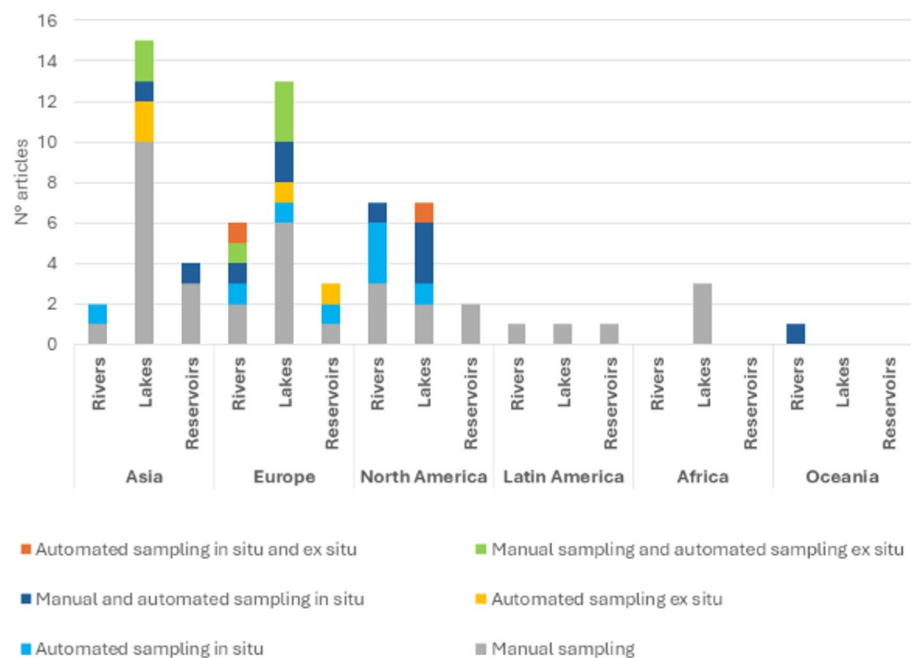
This section presents graphical representations summarising key characteristics of the dataset. First, Table 1 examines the geographical distribution of the case studies contained in the scientific articles. In this review, the vast majority of the identified articles refer to case studies conducted in a specific country, or sometimes to several case studies (generally on the same continent). Only four articles were theoretical and did not involve practical fieldwork. Second, Fig. 2 presents continent-specific approaches to water quality monitoring of surface water bodies presented in the articles.

Although water quality monitoring involves diverse and complementary approaches, a striking pattern emerged: studies focused on (or conducted in) Oceania, Africa and Latin America are under-represented compared with those from Asia, Europe and North America (Table 1b). Table 1a shows that one third of all reviewed articles concern China or the United States, while Asia leads in the total number of publications, followed by Europe (with Germany contributing four articles). This geographic imbalance suggests that current knowledge of water quality monitoring is disproportionately shaped by regions with greater technical and institutional capacity. As a result, monitoring approaches promoted in the literature may not fully reflect the operational realities of emerging economies, where financial, logistical and governance constraints influence technology adoption. Latin America is represented by only three articles (Brazil, Ecuador and Argentina), Africa by three (Ethiopia and Egypt) and Oceania by one (Australia). None of the articles specifically examined sensor development.

With regard to water quality monitoring systems, as far as this specific review report, manual sampling remains predominant across all continents and is the only technique reported in Latin America and Africa (Fig. 2). Indeed, probably many experiences of implementing low-cost devices in developing countries are not even documented in

Table 1 Geographical distribution of case studies in selected articles by country (a) and continent (b)

*The number of cases by country don't include articles with more than 1 country studied

**Fig. 2** Continent-specific approaches to water monitoring

Web of Science. Autonomous systems are widely used in Europe, North America and Asia, and are more developed for rivers than for lakes and reservoirs. Remote sensing is used in lakes, particularly in Europe and Asia, but remains less widely applied in rivers than *in-situ* sensors.

None of the reviewed articles related to in resource-constrained regions focused on sensor development. The widespread availability of low-cost sensors and microcontrollers is transforming monitoring practices; therefore, this imbalance is noteworthy given the importance of water quality monitoring for local policy agendas, as outlined in the introduction of this article. Nevertheless, the close reading of the available literature provides important insights, detailed in the following sections.

3.2 Traditional tools used for monitoring vital signs of surface waters

3.2.1 *Traditional sampling to identify key correlations between the main water quality indicators for calibration purposes*

In the reviewed case studies, water quality monitoring systems based on traditional sampling methods confirm their usefulness and effectiveness in many respects. Water quality parameters are recognised as indicators of aquatic ecosystems's ecological health [9, 14, 41, 62, 72, 77, 113]. Measurement campaigns carried out on a wide range of variables at the start of the study make it possible to determine the key water quality indicators specific to each waterbody that locally influence eutrophication and turbidity processes. Cross-analysis of the results of these measurements also helps to identify correlations, which provide an understanding of the complexity of interactions between physical–chemical and/or biological variables in the degradation of water quality. Jia et al. [43] and Lin and Li [52] reported that turbidity and eutrophication of water bodies were correlated, with a system's sensitivity to nutrient depending on internal factors (resuspension of sediments) and external factors (upstream water discharges). Similarly, Abalasei et al. [1] demonstrated a relationship between turbidity and organic matter in their analysis of Chirita Lake's water quality data, which could contribute to the sustainable management of drinking water in Romania. For their part, Mequanent et al. [56] showed a positive correlation between turbidity and: 1) low vegetation cover in the buffer zone, 2) % of human disturbance assessment and 3) total nitrogen. Identifying these associations is important to long-term monitoring, making it possible to better calibrate predictive models.

Dissolved oxygen emerges as a key variable for assessing rivers, lakes or reservoirs integrity, as its near-bottom depletion is closely linked to eutrophication processes, thermal stratification and the accumulation of organic matter [2]. The spatial and temporal dynamics of dissolved oxygen, particularly in deeper or poorly mixed zones, can exhibit pronounced variability that is not captured by low-frequency or point-based measurements, underscoring the value of continuous and sensor-based monitoring approaches [2]. Therefore, long-term monitoring of each body of water is essential to understand the interactions that influence water quality throughout the year. [49, 87]. At the same time, the precise spatial resolution of this type of measurement is necessary to calibrate remote sensors and *in-situ* loggers to determine the strategic locations for their installation in the basin or water body.

3.2.2 The benefits and complementarity of in-situ and ex-situ sensors

Satellite-based sensors are widely used in several research areas. Yet, as Lepistö et al. [49] highlight, satellite-based remote sensing offers high spatial and temporal resolution, but it needs to be calibrated using field campaign measurements. Caroni et al. [17] used global satellite observations to study 307 shallow lakes (< 3m depth) worldwide over the last 20 years, assessing trends of water quality as chlorophyll-a and turbidity – related to phytoplankton biomass and suspended sediments, respectively. Their study revealed the benefits of low-cost, large spatial coverage and high spatial resolution for water monitoring. The performance of satellite-based techniques varies. Tavares et al. [94] used satellite imagery (Sentinel-2 MSI Imagery) for monitoring superficial events in small lakes (<10 km²) located in the Adour Garonne river basin, calibrated with more than 600 *in-situ* data from 108 lakes. Laneve et al. [47] compared different methods using Sentinel-2 and Landsat 5.7-8, implementing atmospheric corrections in 4 lakes in Italy, obtaining high performances (RMSE close to 3 mg/m³) in lakes with low turbidity. Kiwanuka et al. [46] developed predictive models to improve the health of lakes, based on the experience using Landsat 8 and 9 imagery and *in-situ* samples. In contrast, Mequanent et al [56] used remote sensing, GIS, human disturbance assessment and manual sampling in four sites with varying degrees of alteration in the Lake Tana buffer zone in Ethiopia. Their study showed the value of cross-referencing the accuracy of data measured using a multiparameter probe and a Secchi disk –following Tantu et al.'s [93] approach in lake Abaya in Ethiopia– with data from *ex situ* sensors to better understand interactions between land use and water quality.

In addition, several studies have developed different on-site monitoring systems that yield valuable insights in aquatic ecosystems. García-Avila et al. [31], for example, assessed eutrophication levels in a high Andean lake in Ecuador using multiple trophic indices (OECD, Carlson, and others). Similarly, Prasetyo et al. [70] evaluated water quality in Lake Batur, Indonesia, through a Water Quality Index based on seven parameters, identifying the suitability of water for different uses. More recently, machine-learning approaches have proven effective for calibrating water-quality models, suggesting a strong capacity to identify relationships among *in-situ* parameters [37, 54] and [75]).

Finally, manual data help to programme hydrological models, which are used to predict water quality. In Canada, Zolfaghari et al. [113] employed a combination of modelling, optical sensors and laboratory analyses to study variables such as chlorophyll-a, turbidity and dissolved organic carbon, illustrating the ecological and socio-economic risks associated with high turbidity. Using a quantitative approach, Jamal et al. [41] examined how land use affects the water quality of Wular Lake in the Kashmir Valley, India, based on manual sampling of 22 physicochemical parameters. The analyses led to the design of strategies aiming to protect both the wetland ecosystem and the health of the water body.

Field campaigns remain a standard approach, often requiring extensive logistics, personnel and equipment. Although per-sample costs can be moderate in well-resourced programmes, laboratory analyses are often prohibitively expensive for citizen initiatives. In contrast, the deployment of *in-situ* sensors enables continuous monitoring of water quality parameters, offering temporal precision that traditional field campaigns often lack.

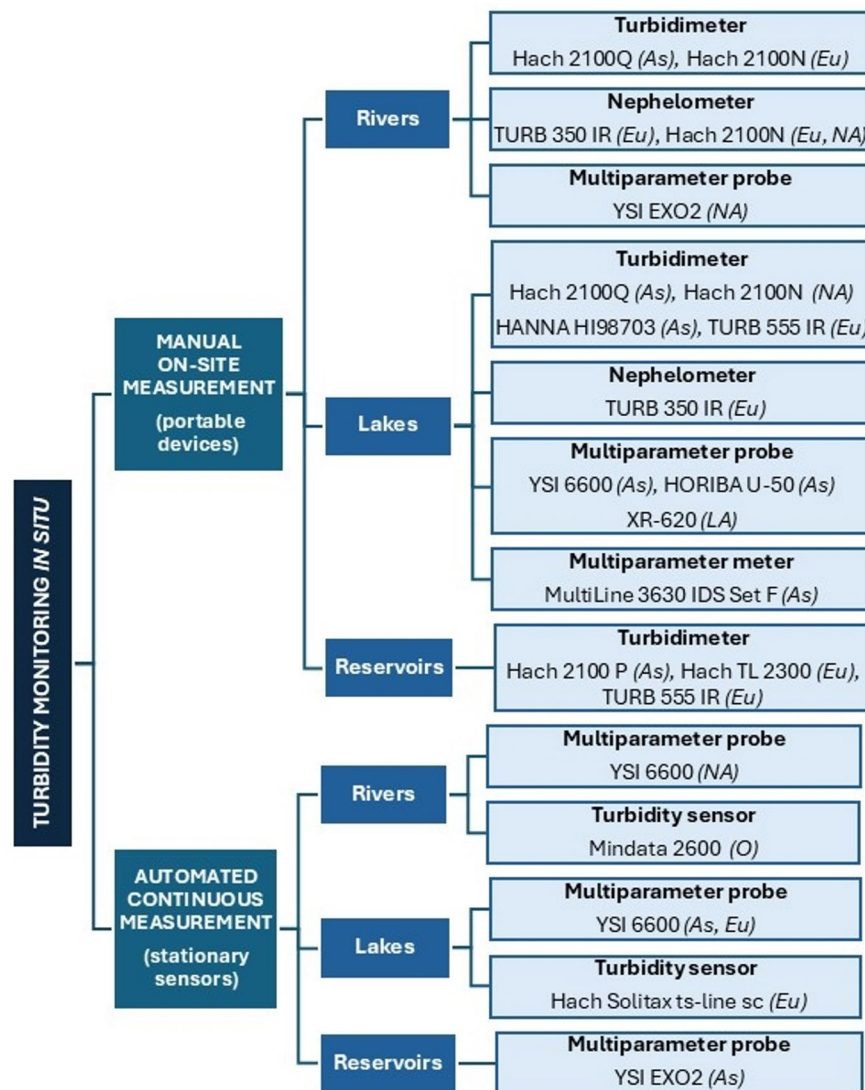


Fig. 3 Types of turbidity sensors for in-situ monitoring. The abbreviations in parentheses refer to the continents where the sensors are used: As for Asia, Eu for Europe, NA for North America, LA for Latin America and O for Oceania

3.3 International experiences in the use of autonomous and continuous sensors, with emphasis on turbidity sensors

In-situ turbidity monitoring relies on the temporary or continuous deployment of instruments in surface waters. This literature review reports on the use of turbidimeters, nephelometers, multiparameter and multimeter probes for manual sampling, and multiparameter probes and turbidity sensors for automated sampling (Fig. 3). Some equipment is used for all types of monitoring.

Among the studies we reviewed, Skarbøvik et al. [87] provided the most comprehensive comparison between *in-situ* turbidity sensors and existing suspended sediments concentration northwestern European streams. The relevance of this study is underscored by the Water Framework Directive, introduced by the European Commission in 2000, which aims to achieve good water quality status across member states. As noted by the authors, optical turbidity sensors offered a promising solution for monitoring suspended sediments concentrations. However, their study also highlighted two limitations

in the current continental monitoring system. First, such sensors are not yet integrated into the national monitoring programmes of Denmark, Finland, Scotland and Northern Ireland, where only data from research projects were available. Second, Sweden, Norway and Ireland have only relatively recently adopted turbidity sensors –since 2017, 2013 and 2009 respectively– making long-term temporal comparisons difficult. These findings align with those of O’Flynn et al. [62], who emphasised the need for a cost-effective, real-time monitoring framework to support the Water Framework Directive objectives, noting its potential scalability to other regions, such as Latin America.

3.4 Towards more effective hydrological models using data from affordable, remote and autonomous sensors for measuring water turbidity

Excess sediment and nutrient loads can pose serious challenges for aquaculture, tourism and public health [4, 21]. Furthermore, factors such as land cover, topography and catchment size, along with meteorological data, can be integrated into semi-distributed hydrologic models [65, 95]. Extending this line of inquiry, several studies have demonstrated that hydrological modelling, when combined with variables like sediment and nutrient concentrations in the water column, is a widely used and effective approach [34, 63, 71, 110]. However, high-frequency sampling –ranging from minutes to daily intervals– over extended periods (e.g., 2 to 7 years) is essential for reliable model calibration and validation. Therefore, acquiring such data in the most cost-effective manner is crucial [6, 63].

In parallel, advanced data fusion techniques, including machine-learning algorithms, can be employed to integrate diverse datasets, allowing for a more nuanced understanding of the relationships between variables and improving the accuracy of model predictions. According to Arnold et al. [7], the widely used Soil and Water Assessment Tool (SWAT) is a simple yet reliable model for simulating hydrological fluxes alongside water quality variables. Similar findings were reported by Yeo et al. [110] and Arabi et al. [5], who used SWAT to simulate water flow, soil erosion, SS concentration and nutrient transport at the catchment scale, as well as to assess the impacts of land-use change, best management practices and climate change.

In contrast, Lepistö et al. [49] used a multi-technical approach to monitor water quality in Pyhajarvi Lake, southwest Finland, using remote sensing data from the Medium Resolution Imaging Spectrometer (MERIS) and transect measurements collected from a moving boat. Their methodology included manual water sampling and laboratory analysis to compare the results with remote sensing data. Although their findings suggest that transect measurements provide accurate estimates of water quality parameters, the costs of measurements remain high, and remote sensing techniques require improvement for certain estimations.

3.5 AI-assisted sensor fusion and cloud-based environmental monitoring as tools that enhance data reliability and accessibility

Beyond data acquisition, several recent studies focused on how large monitoring datasets can be analysed and translated into management decisions. The latter has introduced new methodologies such as hybrid machine-learning models. Gandhimathi et al. [30] integrated regression and boosting techniques to improve river water quality classification, demonstrating the downstream analytical potential of monitoring data. Other

studies have emphasised the use of monitoring data as direct inputs for decision-support and monitoring network design. Batina and Šiljeg [11], for example, combined in-situ water quality measurements with a GIS-based multi-criteria decision analysis framework to optimise the spatial configuration of monitoring stations in a coastal shallow lake. By integrating physicochemical parameters, anthropogenic pressures and meteorological drivers, their approach illustrated that water quality monitoring systems can indeed support early-warning capabilities and management decisions under practical resource constraints.

Based on experiences in various countries, the reviewed literature supports the use of water quality monitoring systems for local-level decision-making. Moreover, the installation of networked sensors offers clear economic advantages. Postolache et al. [69] introduced a low-cost wireless system based on ZigBee technology to monitor surface water quality, highlighting its potential for scalable deployment. In this regard, the co-design of low-cost underwater wireless sensor networks has proved to be a viable alternative for enhancing continuous monitoring systems without incurring the prohibitive costs associated with conventional commercial stations [105]. Another example is the use of high-frequency monitoring buoys, as implemented by Stevens et al. [90] in Lake Erie, North America, which provide a real-time application network for environmental parameters (e.g., temperature, dissolved oxygen, chlorophyll-a, phycocyanin and turbidity). In aquaculture systems, the use of sensors also represents a cost-effective solution for supplying water quality monitoring systems. Petrea et al. [67] applied machine-learning techniques to develop algorithms embedded in soft sensors (models complementing physical sensors), allowing the estimation of parameters such as pH, temperature, ammonia, enabling the prediction of nitrate events in aquaponics systems.

Machine-learning techniques can partially compensate for the lack of continuous *in-situ* turbidity monitoring by reconstructing long-term satellite-derived products. Zou et al. [114] reconstructed a continuous turbidity time series (2004–2021) for Tonle Sap Lake by combining MODIS imagery with Random Forest and XGBoost models to fill multi-year data gaps in Copernicus products. Their results revealed strong seasonal and spatial patterns in turbidity, along with an overall increasing trend, indicating that anthropogenic factors exerted a stronger influence on turbidity dynamics than natural drivers or dam construction. Although this approach does not provide real-time information, it underscored the potential of data-driven methods to recover long-term turbidity dynamics in systems where permanent monitoring infrastructure is limited and can serve to support autonomous *in-situ* monitoring strategies.

3.6 Recommendations for implementing autonomous continuous sensors as an early warning system for real-time decision-making

In recent years, the availability of low-cost sensors and microprocessors at moderate prices has revolutionised field science and environmental observations, democratising access to data collection and analysis [32, 40, 105]. It has facilitated the modelling and integration of various variables that were previously costly and challenging to measure [73]. As noted by Borah et al. [15] and Rollo and Robin [81], model calibration and validation performance are evaluated through statistical metrics. Once validated, such models can be used for scenario analysis, forecasting the impacts of different

environmental changes on turbidity levels, with acknowledging the limitations of low-cost optical telemetry.

Three factors are widely recognised as having the potential to improve future water quality measurement systems: turbidity as a key parameter, the use of optical sensors and the temporal continuity of measurements. First, according to Robertson et al. [80], turbidity is the most cost-effective variable to measure. Integrating hydrological models with turbidity data can help predict when and under what conditions turbidity events may occur. As demonstrated by Rele et al. [77], combining low-cost sensor data with public databases can further enhance the prediction of turbidity in riverine systems.

Second, while Sirabahenda et al. [86] argue that optical sensors should be used to monitor changes in turbidity to ensure accurate water quality assessment, Lepistö et al. [49] emphasise that the resolution offered by these sensors is superior to the imperfect results obtained from interpreting satellite imagery, even when remote sensing is combined with manual field measurements using optical and electrochemical sensors.

Third, as suggested by Zeng et al. [111], high-frequency and continuous time series of measurements seem essential to overcome the biases associated with seasonal variability of water quality parameters. To ensure the reliability of model predictions, long-term time series (exceeding three years) are required, as Robertson et al. [80] demonstrated in their load estimation using two regression models developed by combining the surrogate model (continuous flow, seasonality and surrogates) and the UV-flow model (continuous Unit-Value flow and a seasonality factor). This study offers an interesting option for providing baseline information on water quality and loads, as well as allowing the calculation of historical and real-time daily loading. Similar findings were reported by de Vicente et al. [24], who showed that multi-temporal variability (inter-annual, seasonal and daily) in episodes of water quality deterioration, together with the unpredictability of their drivers –particularly in shallow lakes– highlights the need to increase the frequency of data collection to obtain a more comprehensive long-term dataset. This suggests that a sensor capable of capturing the entire water column would be very useful. However, given the challenges of maintenance and reliance on costly fixed infrastructure, it would be better to develop monitoring based on wireless sensor systems.

4 Discussion

4.1 Clean water is of vital importance

Monitoring the quality of hydrosystems is essential. Favouring the concept of ‘clean water scarcity’ (using both quantitative and qualitative criteria) over that of ‘classic water scarcity’ (using quantitative criteria alone) is projected to triple the number of river basins at risk worldwide between 2010 and 2050 [104]. About 40% of South America’s population could be living in a ‘clean water scarcity’ hotspot by 2050 (under the SSP5—RCP8p5 scenario) [104]. Synthetic fertilisers and domestic waste are the main contributors to river pollution, due to the lack of wastewater collection networks and treatment infrastructures. Furthermore, from an economic perspective, Li et al. [51] have demonstrated that water pollution will affect the economic resilience of foreign trade, as is currently happening in China’s industrialised provinces where productive activities are altering water quality, sometimes irreversibly. Indeed, by 2017, estimated economic losses due to the risk of water shortages had already doubled when water pollution was

included in the evaluation criteria, with economic activities located downstream of catchment areas being the most affected.

4.2 The development of sensors and monitoring networks in Latin America

There is an urgent need for a robust assessment of pressures on aquatic ecosystems and for tools that democratise water quality information for bodies of water – this gap directly impacts societies. Given the limited understanding of the complex interactions between biological, physical, chemical, climatic and hydrological processes across time and space [1], a reliable assessment of water status is therefore of great importance. Identifying patterns of water quality variability can provide surface waters managers, local governments and scientists with valuable insights into the impacts of various human-driven stressors [89]. The degradation of lakes, rivers and reservoirs worldwide threatens the maintenance of essential ecosystem services. In that sense, Latin America is a key stakeholder with strong potential and growing interest in the development of public monitoring of its waterbodies.

The review revealed that acoustic and optical sensors are important for measuring water quality. On one hand, acoustic sensors often work in environments with high particle density and low visibility, where light-based sensors might falter. The use of sound waves allows penetration through turbid water, enabling measurements even in challenging conditions. Their robustness makes them suitable for deployment in harsh environments where durability is essential. On the other hand, optical sensors, such as those based on Nephelometric Turbidity Units (NTU) and Formazin Attenuation Units (FAU), leverage light to measure turbidity [40]. Optical sensors are highly sensitive to small particles, providing accurate data that are crucial for detailed water quality assessment. One significant benefit of optical sensors over acoustic ones under specific environmental conditions is their cost-effectiveness, making them accessible for widespread use. Additionally, their precision and sensitivity make them particularly suitable for quality monitoring, where detailed and accurate data is paramount. Both sensor types have their place in water quality measurement. Optical sensors tend to be favoured for their high-resolution capabilities and affordability. In Latin America, where there is generally a lack of water quality information, combining both sensor types can offer a comprehensive approach to water quality monitoring, supporting the adoption of low-cost yet reliable equipment. Hydrological modeling serves as another critical tool for effective water quality management, as proposed by Arnold et al. [7], by simulating water quality fluctuations driven by hydrological processes at the catchment scale. These models can simulate sediment transport and localised pollution by incorporating diverse point and non-point sources into the equation. Furthermore, the integration of machine-learning with continuous monitoring—using *in-situ* sensors or satellite remote sensing data—has enhanced measurement precision.

The reviewed literature suggests that there is a need to increase the capacity to monitor hydrosystems in different countries, taking advantage of more affordable equipment available on the market. Skarbøvik et al. [87] indicate that more harmonised methods are needed in monitoring programmes in northern Europe, including the development of coordinated measurement plans for various aquatic ecosystems and the integration of results into data networks that allow spatial comparisons of water quality variability. In contrast, the few cases presented in this review that concern Latin America tend to

relate to isolated measurements that are not yet systematically even linked to ongoing monitoring programmes.

Political measures, such as the Water Framework Directive (WFD; 2000/60/EC) applied by the European Commission, promote the monitoring of watercourses, increasing the amount of information available. National or regional monitoring programmes and policies are tools for decision-making and the mitigation and restoration of water quality in lakes and rivers. In Latin America, as shown in this review, multiple bodies of water show signs of deterioration. These cases indicate the need for measures to improve water quality in order to help to prevent contamination, turbidity and eutrophication.

4.3 Connecting citizen science and easy-to-use autonomous sensor networks for widespread water quality monitoring in Latin America

Several factors have influenced the growing number of citizen science initiatives focused on water quality monitoring since 2014 [61]. On one hand, institutional vulnerabilities, high financial constraints and the rigidity of regional discharge legislations may explain the need to produce water quality data from local communities themselves using low-cost devices. On the other hand, this phenomenon could be linked to the rise of low-cost emerging technologies and the growing interest in this field [61]. The authors highlighted the challenge of implementing costly public water monitoring in low-income countries, for which citizen monitoring represents an interesting alternative. However, citizen science in hydrology appears more frequently documented in the Global North (North America and Europe) than in the Global South [27, 61]. This geographical imbalance represents a fundamental diagnostic finding of our review: the under-representation of Latin American literature in major indexing databases does not reflect an absence of water quality crises, but rather a structural asymmetry in research funding and technological adoption. Unlike in the Global North, where data collected through participatory science is sometimes integrated by the authorities, citizen water monitoring in the Global South is frequently carried out more to defend territories exposed to threats and to challenge public authorities, which, however, do not universally recognise these measurements [27]. In Latin America, this asymmetric relationship is aggravated by obsolete regulatory frameworks. For instance, the systematic delay in updating key environmental standards—such as Decree D.S. N°90/2000 in Chile—creates an institutional vacuum where modern parameters of ecological integrity are ignored in official enforcement, forcing local populations to operate under high financial constraints to generate their own defensive environmental data.

Extractive industrial activities, especially mining, are the main concerns behind citizen water initiatives in Latin America [27]. The presence of cyanobacteria associated with nutrient enrichment and increased water turbidity in lakes is another source of interest [29]. In contrast, despite expanding rural change processes, concerns over pollution risks to drinking water and recreational surface waters remain limited, particularly from direct wastewater discharge or septic tanks [8].

These projects primarily focus on measuring surface water and collecting chemical, physical and biological data [27]. The main parameters measured are clarity, nitrate, pH, dissolved oxygen, phosphate, temperature, macroinvertebrates, electrical conductivity and faecal coliform bacteria (*E. coli*) [13]. Strip-type colorimetric tests are the most widely used despite their lack of accuracy, followed by the Secchi disk for turbidity and

a series of test kits approved by accredited organisations [13]. These initiatives rely on manual sampling campaigns, which are often sporadic and constrained by the high cost of laboratory analyses. Consequently, these localized networks currently function as reactive, point-in-time responses to environmental contamination. The integration of low-cost continuous telemetry into these existing Latin American networks holds the potential to structurally transform these sporadic efforts. By overcoming the economic barrier of laboratory analysis and traditional station maintenance, continuous telemetry can elevate community-led monitoring into automated, high-frequency early-warning systems capable of capturing sudden eutrophication and turbidity events that currently escape both official state surveillance and intermittent citizen campaigns. However, [61] qualify that training and certification programmes make it possible to produce relatively reliable databases. The development of low-cost methods based on camera or smart-phone-based systems and wireless sensor networks, such as Zigbee, seems promising for promoting continuous water citizen science in territories facing water quality challenges [61]. Therefore, citizen science networks are increasingly complementing government monitoring programs by addressing data deficiencies, and the integration of low-cost continuous telemetry could elevate these sporadic observations into systematic and reliable early-warning systems within the limits of low-cost optical telemetry.

4.4 Common way forward

Across the reviewed studies, and although some articles may have been inadvertently omitted, it became clear that maintaining water quality requires a multifaceted approach combining advanced monitoring technologies, hydrological modelling and proactive management strategies. Although challenging, a widely supported path forward involves integrating low-cost, autonomous sensor networks with traditional monitoring methods and remote sensing techniques. This integrated approach presents a range of advantages and limitations that merit careful consideration.

However, moving beyond a descriptive synthesis requires a critical evaluation of the socio-technical friction points inherent to these hybrid frameworks. The transition toward low-cost environmental observation is not merely a matter of technological substitution, but a structural shift in how environmental uncertainty is managed and validated.

4.4.1 Advantages

- Availability of high-frequency data: While manual sampling may be limited by seasonal weather constraints [56], autonomous sensors provide high-frequency data, enabling early detection of water quality changes year-round and the development of timely alert systems [49, 113]. This high temporal resolution effectively shifts the paradigm from reactive monitoring to anticipatory limnological management
- Cost-effectiveness: Low-cost sensors reduce reliance on expensive and labour-intensive field campaigns, making extensive monitoring—especially in remote or under-resourced regions—more feasible [69].
- Enhanced decision-making: Real-time data integrated into hydrological models supports informed management and regulatory decisions. It also opens opportunities for citizen science initiatives that can empower local communities.

- Scalability and flexibility: Combining sensor data with remote sensing and modelling enables assessments over wide spatial and temporal scales, essential for developing sustainable public policies and legislation [1] to address both local and global water quality challenges.

4.4.2 Limitations:

- Calibration and data quality issues: Despite their advantages, low-cost sensors require regular calibration and validation to ensure data accuracy, given seasonal variations and spatial heterogeneity [49, 87], before they can be used for robust hydrological models, for example. Their wider adoption also depends on demonstrating performance and quantifying measurement uncertainty, as they are often perceived as less reliable than conventional monitoring methods. Standardized protocols, such as EN 17075+A1:2023 and ISO 15839:2005, provide frameworks for evaluating sensor performance under real-world conditions. From a critical standpoint, the trade-off of spatial flexibility in low-cost arrays is a continuous vulnerability to operational drift. Biological fouling—such as biofilm accretion on optical windows—and lens sedimentation cause progressive signal attenuation that requires systematic, often resource-intensive field maintenance, partially offsetting the initial economic advantage of these devices.
- Technical and logistical challenges: Deploying and maintaining autonomous sensor networks can be challenging, particularly in harsh or remote environments. Challenges include ensuring continuous power supply, data transmission integrity and sensor longevity. In addition, integrating data from various sources (manual sampling, satellite imagery, *in-situ* sensors) into unified, robust hydrological models is complex and demands significant technical expertise, which is frequently lacking. Moreover, prolonged use of the sensors leads to biological fouling, which causes progressive data drift and signal attenuation. Therefore, to ensure the reliability and continuity of autonomous systems, they should be equipped with antifouling mechanisms, such as wipers or compressed air purges.
- Policy and coordination needs: While the successful implementation of these technologies requires updated technical guidelines and the coordination of multiple stakeholders, including water utilities, local governments and research institutions, systemic institutional barriers continue to limit the integration of community-validated citizen science data into official regulatory frameworks.

A review of this literature provides a deeper understanding that the most promising strategy is to use the complementary strengths of traditional and emerging water quality monitoring systems, for example, training local communities to employ low-cost sensors. Nevertheless, several challenges need to be addressed, such as sensor calibration, ensuring robust data integration and nurturing collaborative governance frameworks. To enhance the long-term reliability and interoperability of low-cost sensor networks, future implementations should incorporate regular validation procedures, rigorous site-specific calibration protocols and maintenance strategies to mitigate the effects of biofouling, lens sedimentation and seasonal temperature variability. In addition, the development of harmonised data-processing frameworks is recommended to improve data comparability and facilitate the integration of citizen-generated observations into

regional and public monitoring networks. By engaging diverse stakeholders, it becomes possible to develop effective early warning systems and sustainable water management practices. This integrated approach not only enhances scientific understanding of water quality dynamics in monitored ecosystems but also equip local decision-makers with actionable insights to safeguard critical hydrosystems amid increasing environmental pressures.

5 Conclusion

Water quality is increasingly threatened by population growth, urbanisation, land-use change and climate-driven pressures. This review has synthesised the core literature on the evolution of water quality monitoring systems applied to aquatic ecosystems affected by turbidity and eutrophication, drawing primarily on peer-reviewed studies indexed in Web of Science. The analysed studies consistently highlight growing environmental and public-health risks associated with declining water quality.

In line with the hypotheses guiding this review, published monitoring practice in lakes and rivers remains dominated by grab sampling campaigns and satellite-based approaches, while continuous autonomous *in-situ* monitoring is comparatively less represented, except in rivers. At the same time, the literature reveals a persistent gap in accessible real-time information, particularly in regions with limited resources, where monitoring often depends on specialised technical capacity and uneven station coverage. Despite the increasing availability of affordable sensors capable of generating high-frequency datasets, their systematic integration into coordinated monitoring networks remains limited. These insights suggest that the principal barrier to improving monitoring is no longer technological availability but the institutional and operational integration of autonomous systems into structured observation frameworks. From a research perspective, the evidence supports deploying sensors both at basin scale and at strategically selected locations within water bodies to capture spatial variability and identify pollution sources. Such configurations can strengthen model calibration, improve early-warning capacity and enhance long-term environmental assessment.

To ensure future water security worldwide, two key public policy measures appear promising. On the one hand, the deployment of sensor networks in surface waters is crucial for preserving ecological processes while supporting socio-economic, productive, recreational and educational activities that depend on reliable water resources. On the other hand, the development of autonomous and participatory monitoring systems involving local communities could significantly improve real-time decision-making and promote more adaptive water governance.

In looking ahead, the decisive question will be whether systems can transform increasingly available high-resolution data into effective environmental action alongside current monitoring frameworks. Global challenges of water quality will thus require not only an effective data-to-decision chain, but also a broader transition from fragmented monitoring architectures towards more coordinated and responsive governance.

Supplementary Information

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Supplementary Material 1

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Author contributions

LLG: Conceptualisation, Methodology, Writing—Original draft preparation, Data curation, Reviewing, Editing, Validation. NF: Conceptualisation, Writing—Original draft preparation, Supervision, Reviewing, Editing. MS: Conceptualisation, Methodology, Writing, Data curation, Editing, Reviewing. PH: Conceptualisation, Writing, Editing, Reviewing. RA: Methodological support, Reviewing, Editing

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Data availability

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Declarations

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